Week 1 lab

1. Question: How does configuring an enable secret password help to protect a router from being compromised by an attack?
    1. configuring an enable secret password prevents users without the password from elevating their access further into enable or configuration mode, this can prevent malicious actors from obtaining or destroying information or inexperienced users accidentally causing damage.
2. Question: When you configured the password for the console line, what message was displayed?
    1. Invalid password length, must contain 10-127 characters. Password configuration failed
3. Questions:
    1. Were you able to login? Explain.
        1. No, we have not set the VTY password yet
    2. What messages were displayed?
        1. password required, but none set
4. Question: Were you able to login this time?
    1. Yes, after setting the line vty password we can now login
5. Question:
    1. Can you read the enable secret password? Explain.
        1. No, the password is currently hashed
    2. Can you read the console, aux, and vty passwords? Explain.
        1. Yes, when the enable password was set we clarified that we wanted to encrypt it and set the encryption type. we did not do that for the other passwords
6. Questions:
    1. Can you read the console, aux, and vty passwords? Explain.
        1. No, the passwords have no been encrypted
    2. At what level (number) is the default enable secret password encrypted?
        1. type 9.
    3. At what level (number) are the other passwords encrypted?
        1. type 7.
    4. Which level of encryption is harder to crack. Explain.
        1. type 9, type 7's purpose is to prevent viewing of passwords via the show run command and is easily broken. type 9 is a more complex hashing algorithm that is more computationally expensive to crack and encrypt
7. Question: What does the $ convert to in the output?
    1. a ^c or a Control + C input

8. Question: What options are available
    1. MD5, Scrypt and sha256
9. Question: Which hashing method is used for the password?
    1. Scrypt
10. Question: What is the difference between logging in at the console now and previously?
    1. you are now asked for a username and password
11. Question: Were you able to issue the command? Explain.
    1. No, as we are not in exec mode
12. Question: Were you prompted for a password? Explain.
    1. yes we are asked for the enable password
13. Question: What was the router prompt after you entered the password?
    1. the router was already in enable mode after logging in as an admin
14. Question: What users are connected to router R1 at this time?
    1. Admin via the console, and admin via SSH

Pass Reflection

1. Explain the importance of securing router access and monitoring network devices.
    1. access should be secured to prevent bad actors from accessing or deleting sensitive data or device configurations, monitoring network devices helps us identify when an access attempt has been made and to identify potential vulnerabilities
2. What advantages does SSH have over Telnet?
    1. The key advantage of SSH is the encryption of data that is transmited using the protocol, this includes the password used to access the switch preventing attackers from using man in the middle attacks or other methods of packet interseption from finding a plain text password.

Credit questions

1. Question: Are you able to establish an SSH session with the switch?
    1. Yes
2. Question: What versions of SSH are supported from the CLI?
    1. both version 2 and 1.99 support CLI connections, cisco devices need to have ssh enabled but will be able to SSH via the CLI once done so

Creddit reflection

1. How would you provide multiple users, each with their own username, access to a network device?
    1. with the tools that have been comunicated to us so far i would suggest making a set of credentials for each user on the local devices data base, SSH V2 can

support multiple shells at the same time so this should not cause issues. in future labs we will learn about AAA servers which serve as a centeralized solution for storing user credentails for access which i would reccomend as a better solution for managing user credentials for SSH access across multiple devices